

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
(CHARUSAT)
CHANGA
FIRST SEMESTER EXAMINATION

NT 703- CONCEPTS OF BIOLOGICAL SCIENCES

Date: 03.12.2009

Total Marks: 70

Time: 10.00 AM to 1.00 PM

[PART A]

Total Marks: 20

Put a tick mark (✓) in front of the correct answer:

1. The temperature at which 50% of the DNA is denatured is termed as
(a) pKa (b) T_m (c) G+C (d) none of the above
2. "cap" is a characteristic feature of
(a) mRNA (b) tRNA (c) rRNA (d) DNA
3. Vitamin E belongs to the following class of lipids
(a) prostaglandins (b) terpenes (c) sphingolipids (d) none of the above
4. The following is a non-polar and aromatic amino acid
(a) tyrosine (b) phenyl alanine (c) proline (d) none of the above
5. In carbohydrates the monosaccharides are linked through
(a) amide bond (b) peptide bond (c) glycosidic bond (d) phosphodiester bond
6. The word cell was coined by
(a) J. D. Watson (b) Theodor Schwann (c) Robert Hook (d) Rudolf Virchow
7. Protein synthesis in a cell takes place in
(a) ribosomes (b) mitochondria (c) chloroplast (d) none of the above
8. To be classified as an enzyme a protein must have
(a) substrate and product specificity (b) remain unchanged by the reaction
(c) accelerate the reaction rate (d) all of the above
9. The movement of solute molecules from a higher concentration to a lower concentration is regarded as
(a) viscosity (b) osmosis (c) diffusion (d) solution

10. Colloidal state is characterized by the particle size of
(a) $< 1\text{ nm}$ (b) $> 1000\text{ nm}$ (c) $1 - 100\text{ nm}$ (d) $> 100\text{ nm}$
11. The force with which the molecules on the surface are held together is called as
(a) viscosity (b) osmosis (c) diffusion (d) surface tension
12. The number of moles of solute per liter of a solution is
(a) normality (b) molarity (c) parts per million (d) percent concentration
13. X-ray diffraction analysis is based on
(a) Beer's Law (b) Bouguer-Lambert law (c) Bragg's law (d) all of the above
14. Chemical shift is expressed in
(a) mg/dl (b) gram percent (c) ppm (d) mole
15. Splitting of the lines in the NMR spectrum is due to
(a) chemical Shift (b) positive shielding (c) stock's shift (d) spin-spin coupling
16. Which of the following methods would be best for determination of three dimensional protein structures?
(a) mass spectrum and UV spectrophotometer (b) NMR and ESR
(c) NMR and X-Ray crystallography (d) none of the above
17. Which of the following compounds is used as the standard to measure the magnitudes of chemical shifts of different kind of protons?
(a) Tetra methyl silane (b) Tetra methyl phosphate (c) BSA (d) DDS
18. Which of the following is **NOT** true for polymers?
(a) macromolecule formed by linking of repeated units
(b) have huge molecular weights
(c) properties are independent of backbone structure
(d) none of the above
19. Polysiloxanes are best described as
(a) reversible polymers (b) condensation polymers
(c) organic polymer (d) inorganic polymer
20. Which step in anionic polymerization involves alkali metals?
(a) initiation (b) propagation (c) termination (d) all of the above

[PART B]

Total marks: 50

(Please write answers for each section two separate answer sheets)

SECTION I

- Q. 1 (A) Write short notes on **any two** of the following: (06)
 (i) viscosity (ii) surface tension
 (iii) osmosis (iv) properties of colloids
- Q. 1 (B) What are colloids? Give a detailed classification of colloids. (04)
- Q.2 (A) Define co-polymers. Explain in detail various types of co-polymers. (03)
- Q.2 (B) Enlist the various types of addition polymerization. Write the reaction mechanism of any one type of addition polymerization. (02)
- Q.2 (C) Explain the phenomenon of spontaneous termination in conventional Polymerization. (02)
- Q.2 (D) Elaborate on the mechanism of Zeigler-Natta catalysis. (03)

SECTION II

- Q.3 (A) Describing salient features of fatty acids, write a short note on phosphoacyl glycerols. (03)

OR

- Q.3 (A) Discuss: Sphingolipids and their functions. (03)
- Q.3 (B) Describe the structure and conformations of DNA molecule. (04)
- Q.3 (C) Write a short note on **any one** of the following: (03)
 (i) structure of eukaryotic mRNA (ii) Denaturation of DNA (iii) steroids
- Q.4 (A) What are enzymes? Discuss: Michelis- Menten equation (04)
- Q.4 (B) Draw a typical mammalian cell. Describe the structure and functions of mitochondria. (03)
- Q.4 (C) Explain **any three** of the following in one or two sentences: (03)
 (i) katala (ii) specific activity (iii) turnover number
 (iv) proteasome (v) chromosomes (vi) cofactors
- Q.5 (A) Give a list of various X-ray diffraction methods. Describe rotating crystals method in detail. (05)

OR

- Q.5 (A) Write briefly on various steps of X-ray structure analysis of protein. (05)
- Q.5 (B) Explain the principle of the NMR spectroscopy. (05)